A cost-efficiency frontier for humanoid robot deployment in parcel sorting

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ABSTRACT
Parcel sorting requires high throughput, routing accuracy, and effective labour allocation as parcel volumes vary across operating periods. Humanoid robots may offer a flexible automation option for facilities designed around human workers, but their economic value depends on sustained throughput, task coverage, and total deployment cost. This paper develops a cost-efficiency frontier for humanoid robot deployment in parcel sorting. It combines human labour costs and robot deployment costs and expresses them as the cost per successfully processed parcel. It also derives unit cost and three-year payback frontiers and formulates a mixed-integer linear programming model to determine the required numbers of workers and humanoid robots. Low-, normal-, and peak-demand conditions are evaluated when only part of the parcel workload can be assigned to humanoid robots. Monte Carlo simulation tests the robustness of the payback results under uncertainty in labour cost, robot throughput, maintenance, and fixed operating costs. The results show that economic viability improves with sustained robot throughput and human labour cost. Humanoid robots are most promising for selective hybrid deployment, where robots process standard parcels and workers manage irregular parcels and operational exceptions.

KEYWORDS
Parcel sorting; Warehouse automation; Humanoid robots; Human-robot collaboration; Cost-efficiency frontier

1. Introduction

Parcel sorting centres must process large parcel volumes within short operating windows while maintaining routing accuracy, service reliability, and cost control (Chen et al., 2024). The process involves repeated handling, parcel orientation, scanning, and transfer to downstream flows. Automation remains difficult because parcels vary in size, weight, shape, packaging condition, and label visibility (Shi et al., 2025). These differences affect grasping, scan success, and transfer performance (Mu et al., 2025).

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Therefore, workers are still needed when the standard flow fails or parcels require local adjustment.

Variations in parcel volume increase operating pressure on parcel sorting centres. Peak shopping periods create short capacity surges that are often met through overtime and temporary labour. Parcel volumes can rise sharply, while manual handling capacity is difficult to expand at short notice. Recruitment difficulties, labour turnover, ageing workforces, and wage growth further increase pressure on sorting operations. Parcel sorting faces three linked pressures, including demand peaks, limited labour availability, and rising labour cost. Warehouse automation already supports many sorting and fulfilment activities. Conveyor and barcode systems provide identification and routing (Boysen et al., 2019). Automated storage systems improve storage and material movement (Azadeh et al., 2019). Robotic mobile fulfilment systems (RMFSs) add mobility and decentralised control (Azadeh et al., 2019). These technologies create complex operational problems in layout, routing, station allocation, fleet sizing, congestion, and control (Boysen et al., 2023; da Costa Barros and Nascimento, 2021; Fang et al., 2025). Their performance also depends on the coordination of physical resources, information flows, station capacity, and downstream policies (see, e.g., Boysen and De Koster, 2025; Zhao et al., 2026).

Existing automation performs best when parcels follow defined routes and handling points are fixed. Manual work remains important in brownfield facilities, mixed parcel flows, and exception processes (Fedtke and Boysen, 2017). Workers reposition parcels, locate labels, complete scans, separate difficult items, and recover damaged or failed parcels (Shi et al., 2025). These tasks require flexibility at interfaces designed around human reach and movement. Humanoid robots may provide this flexibility without a full facility redesign. Their form combines locomotion, manipulation, perception, and access to workspaces designed for people (Rutili de Lima et al., 2024; Tong et al., 2024). In parcel sorting, they may operate around belts, cages, scanners, shelves, and trolleys. Their likely role is selective rather than comprehensive. Robots may process standard parcels, while workers retain responsibility for irregular parcels, damaged packaging, failed scans, supervision, and recovery.

Recent industrial pilots show that humanoid robots are moving from concept development towards practical use. Agility Robotics has reported the use of Digit for tote handling in logistics facilities (Agility Robotics, 2025). Figure has demonstrated package handling through its Helix platform (Figure AI, 2025). Apptronik uses Apollo for industrial case handling and line feeding (Ackerman, 2023). The Unitree G1 provides a relevant benchmark because its publicly available price allows the throughput required for economic viability to be estimated (Unitree Robotics, 2026). These examples demonstrate technical progress but do not establish economic viability in parcel sorting. Practical deployment requires sustained throughput under parcel variation, congestion, downtime, and intervention events (Shi et al., 2025). Warehouse operators must assess task coverage, stoppages, supervision, recovery requirements, and the full cost of hardware, integration, software, maintenance, energy, and downtime (see, e.g., Jacob et al., 2026; Yu and Nayak, 2026).

Existing research covers three related areas. Humanoid robotics focuses on locomotion, balance, manipulation, perception, and task learning (Li et al., 2026b; Tong et al., 2024). Operations research studies on warehousing address parcel sortation, robotic fulfilment, routing, station allocation, scheduling, congestion, and capacity (Boysen et al., 2019, 2023; Teck and Dewil, 2022; Zhao et al., 2025). Research on collaboration between humans and robots considers task allocation, safety, ergonomics, supervision, and worker acceptance (Fragapane et al., 2021; Keshvarparast et al., 2025; Pasparakis et al., 2023; Winkelhaus et al., 2022). These research areas do not yet provide an economic model for evaluating humanoid robots in parcel sorting operations. Existing warehouse models usually represent automation as conveyors, mobile robots, robotic arms, or abstract service resources. Humanoid robots combine mobility, manipulation, perception, and reach in a single platform (Gu et al., 2026). The use of humanoid robots in intralogistics depends on three operating factors: sustained throughput, full deployment cost, and the allocation of parcel workload between robots and workers. This study addresses the following research questions:

- RQ1. What sustained throughput is required for a humanoid robot to match the unit cost of human sorting labour and achieve a target payback period?
- RQ2. How do robot price, human labour cost, assignable workload share, and seasonal operating pressure affect this threshold?
- RQ3. How should work be allocated between robots and workers when robots can process only part of the parcel workload?

To answer these questions, we develop a cost model using public data on labour, robot prices, and logistics operations. Human labour and robot deployment are compared through cost per successfully processed parcel. The model converts labour into annual employer cost and effective output. Robot cost includes acquisition, integration, depreciation, maintenance, software, supervision, electricity, and downtime. Two reference cases from SF Express and JD Logistics, together with the public price of the Unitree G1, are used to derive the cost-efficiency and three-year payback frontiers. The analysis also considers seasonal demand, hybrid staffing, and parameter uncertainty. This study makes three contributions. First, it evaluates humanoid robot deployment through economic thresholds rather than technical capability alone. Second, it links robot price, sustained throughput, human unit cost, and payback within a cost-efficiency frontier. Third, it models the allocation of robots and workers when robot task coverage is limited. The results identify the operating conditions under which humanoid robots can become economically viable parcel sorting resources.

The remainder of the paper is organised as follows. Section 2 reviews the related literature and identifies the research gap. Section 3 presents the data and modelling approach, including the cost, frontier, allocation, and uncertainty. Section 4 reports the numerical results. Section 5 discusses economic readiness and hybrid deployment. Section 6 concludes the paper and presents future research directions.

2. Background and related literature

This section reviews parcel sorting, warehouse automation, and humanoid robots. It examines the development of warehouse automation from fixed systems to mobile robots and identifies the manual handling tasks that may be suitable for humanoid robots.

Parcel sorting has long been automated by reducing process variability. Industrial robotic arms created value in fixed workcells with predefined parts, tools, and motion paths. Conveyor sorters and automated storage and retrieval systems applied the same logic through planned routes, scanning points, storage locations, and destinations (Azadeh et al., 2019; Boysen et al., 2019). Automated guided vehicles (AGVs), autonomous mobile robots (AMRs), RMFSs, and robotic sorting systems then added mobility and decentralised control. Recent research shows that system performance depends on coordinating physical resources, information flows, station capacity, and downstream control policies (Boysen and De Koster, 2025; Boysen et al., 2023; Fang et al., 2025; Zhao et al., 2026). Table 1 summarises the development of warehouse automation from fixed systems to mobile and flexible robots.

Table 1. Timeline of warehousing automation and the emergence of humanoid robots

Period	Automation type	Operational role in parcel logistics
1960s–1980s	Industrial robotic arms and fixed automation cells	Automation expanded in controlled production settings with predefined objects, tools, and motion paths. Robotic arms reduced repetitive work in fixed workcells where items arrived in known positions and orientations.
1980s–2000s	Conveyors, automated storage and retrieval systems, barcode systems and early guided vehicles	Higher warehouse and parcel volumes increased the need for reliable movement and item identification. These systems organised flows around fixed routes, scanning points, storage locations and destination points.
2000s–2010s	AGVs, AMRs, RMFSs, and robotic sorting systems	E-commerce growth and smaller, more varied orders increased the value of mobile and fleet-based automation. Research increasingly focused on routing, dispatching, congestion control, station assignment, and workload balancing.
2010s–early 2020s	Collaborative robots, robotic picking, monitoring systems and activity recognition	Automation moved closer to human workstations through sensing, task support and shared workspace operation. The focus expanded to safety, usability, worker interaction and effective workstation capacity.
Early–mid 2020s	Humanoid robots in demonstrations, pilots and early industrial use	Humanoid robots are being tested for manual handling tasks in facilities built for human workers. They can move between belts, cages, scanners, shelves, and workstations and handle items across these areas.

Humanoid robots address handling tasks that existing logistics automation does not fully cover. Conveyor sorters achieve high throughput when induction points, destination lanes, and downstream flows are predefined (Boysen et al., 2019). Robotic sorting systems and AMR fleets bring routing flexibility and are commonly studied through assignment, congestion control, and dynamic routing models (Boysen et al., 2023; Fang et al., 2025; Zhao et al., 2026). Robotic arms and compact sorter modules extend automation to smaller facilities when item position, workcell layout, and handling sequences are tightly controlled (Li et al., 2026a). These technologies perform best when the process is designed around the equipment. Brownfield facilities present a different challenge. Parcels vary in size and condition, labels are not always placed consistently, and workers often make local adjustments during sorting. Humanoid robots may be suitable for these facilities because they are designed to operate in spaces and around equipment intended for human workers. Their role is likely to remain selective. Robots may process standard parcels, while workers handle damaged items, irregular packaging, supervision, and recovery. Economic viability therefore depends on whether robots can sustain sufficient throughput, process a large enough share of the parcel mix, limit human intervention, and recover their deployment cost. Industrial examples show growing interest in humanoid robots for logistics handling tasks, although most applications remain at the demonstration, pilot, or early deployment stage. Their use in parcel sorting must therefore be assessed in terms of technical capability, operating performance, and economic viability. Table 2 summarises current developments. The classifications are based on publicly reported company announcements, product information, and industrial pilot reports available up to July 2026.

The most relevant industrial applications are found in warehouse handling, parcel manipulation, and manufacturing support. These settings combine repetitive physical work with equipment and workspaces designed for people. Parcel sorting centres

Table 2. Humanoid and related mobile manipulation platforms with potential logistics applications

Country	Companies and platforms	Robot type	Applications	Stage
Canada	Sanctuary AI: Phoenix	Bipedal humanoid with dexterous manipulation	General purpose work, industrial service and manipulation	Commercial pilot
China	Unitree: G1, H1, R1, H2; AgiBot; UBTECH Walker; Fourier GR; Kepler Forerunner; EngineAI; XPeng Iron; Xiaomi CyberOne; RobotEra STAR1; Galaxea; Galbot; Astribot; Dobot Atom; Deep Robotics DR02; Leju; Booster	Bipedal humanoids; hybrid wheeled and bipedal systems; upper body manipulators	Manufacturing, automotive, warehouse trials, logistics handling, inspection and developer platforms	Commercial products, industrial pilots and R&D
France	Pollen Reachy 2; Wandercraft Atalante; Aldebaran NAO	Upper body humanoid, and assistive biped	Manipulation research, assistive robotics and industrial technology transfer	Commercial products, industrial pilots and R&D
Germany	NEURA 4NE-1; Agile Robots DIANA	Bipedal humanoid; force-controlled manipulator; collaborative robot	Manufacturing, industrial integration, manipulation and shared workspaces	Pre-commercial and early production
India	Addverb; Gridbots; Invento Mitra; Asimov Robotics	Warehouse robots; service humanoids; unmanned systems	Warehouse automation, AMRs, sortation, industrial robotics and service operations	Commercial products and pilot deployment
Italy	Oversonic RoBee; iCub-related platforms	Bipedal	Service industry deployment, manufacturing support and cognitive robotics	Commercial pilot and R&D
Japan	Kawasaki Kaleido; Kawada NEXTAGE; Gitai; A-Lab; PaXini TORA-ONE	Bipedal humanoids; dual arm systems; exoskeletons; tactile platforms	Industrial robotics, flexible assembly, support systems, disaster response and parcel and material handling tasks	Commercial products, pilot deployment, and R&D
Norway	1X NEO; Halodi EVE	Bipedal service humanoids designed for workspaces used by people	Industrial assistance, facility support and supervised service work	Commercial deployment
Singapore	Botsync	AMRs and warehouse robots	Manufacturing logistics, warehouse transport and factory material movement	Commercial deployment
South Korea	Robotis DARWIN-OP; Rainbow Robotics; LG CLOiD; Samsung Bot Handy and Ballie; Naver AMBIDEX	Bipedal humanoids; wheeled humanoids; upper body manipulators; service robots	Electronics manufacturing, smart buildings, service robotics and parcel and material handling tasks	Commercial products and R&D
Spain	PAL Robotics TALOS	Bipedal research humanoid	Manipulation, locomotion and industrial robotics research	Commercial research platform and R&D
United States	Agility Digit; Figure 02 and 03; Apptronik Apollo; Tesla Optimus; Boston Dynamics Atlas; Collaborative Robotics; Persona AI; Reflex; HEBI	Bipedal humanoids; mobile manipulators; exoskeletons; assistive systems	Warehouse logistics, tote handling, parcel manipulation, case handling, manufacturing and industrial support	Commercial deployment, industrial pilots and R&D
Vietnam	VinMotion Motion1 and Motion2	Humanoid robots	Automotive manufacturing and factory deployment	Prototype and industrial pilot

may therefore offer suitable conditions for humanoid robots, as they already contain belts, cages, scanners, trolleys, shelves, workstations, and walkways intended for human use. Robot design affects both task coverage and operating cost. Bipedal robots may adapt more easily to existing facilities, while wheeled or upper body systems may offer greater stability at a lower cost. Dexterous robots may handle a wider range of parcels, while systems that are easier to integrate may reduce implementation costs. Economic viability depends on robot price, deployment cost, task coverage, reliability, and sustained throughput. Recent humanoid robotics research focuses on locomotion, balance, whole-body control, manipulation, perception, and task learning (Li et al., 2026b; Tong et al., 2024). Studies of Atlas, DURUS, REEM-C, and Duke Humanoid examine planning, optimal control, model predictive control, and reinforcement learning for stable and energy-efficient motion (Hereid et al., 2018; Kuindersma et al., 2016). Related work on motion retargeting, human motion reconstruction, and task planning shows how human demonstrations can be converted into executable humanoid actions (Ayusawa and Yoshida, 2017; Jacob et al., 2026).

Warehouse research provides the models and algorithms needed to evaluate how robot capability affects operating performance. Studies of robotised warehouses and RMFS examine layout, pod selection, station allocation, sequencing, routing, congestion control, and demand uncertainty (Kiabakht and Sajadieh, 2026; Teck and Dewil, 2022; Zhao et al., 2025). Research on human and robot collaboration addresses safety, ergonomics, task allocation, supervision, fatigue, acceptance, blocking, and implementation conditions in mixed work systems (Fragapane et al., 2021; Keshvarparast et al., 2025; Ma et al., 2026; Pasparakis et al., 2023; Winkelhaus et al., 2022). Data-driven and queueing studies estimate activity times, service rates, buffer requirements, congestion, downtime, utilisation, and effective capacity (Amjath et al., 2024; Fedtke and Boysen, 2017; Zhang et al., 2024). Together, these studies explain how warehouse systems are designed, measured, and controlled. However, robots are usually modelled as conveyors, mobile carriers, fulfilment robots, robotic arms, or service resources. Full-body humanoids rarely appear as explicit economic resources in parcel sorting models. Table 3 summarises these research areas and their relevance to the present study.

The covered literature provides limited guidance on the economic evaluation of humanoid robots for parcel sorting. Humanoid robotics focuses on technical capability, warehouse research on operating performance, and human-robot collaboration on shared work. However, few studies model a humanoid robot as a parcel sorting resource with acquisition cost, deployment and support requirements, throughput, reliability, and limited task coverage. This study addresses that gap by deriving the throughput required to match human unit cost and achieve a target payback period. It also evaluates hybrid staffing when robots can process only part of the parcel workload.

3. Methodology

This section presents the data, assumptions, and models used to evaluate the economic viability of humanoid robots in parcel sorting. Parcel sorting relies on repeated manual handling because of parcel variability, facility design, and exception processes. Therefore, the economic value of a humanoid robot depends on whether it can replace enough of this work at a sufficient and reliable throughput to justify its deployment cost. Two aspects are important when evaluating humanoid robots for parcel sorting. These include task substitution and operational flexibility. Task substitution is the share of parcel handling work that can be transferred from workers to robots. Operational flexibility is the ability of a robot to work within an existing sorting facility without extensive redesign. Conventional systems such as conveyors, cross-belt sorters, and automated sorting equipment can achieve high throughput, but their operation is closely linked to facility layout, parcel flow, and operating rules (Zou et al., 2021). Hu-

Table 3. Research areas relevant to humanoid robot deployment in parcel sorting

Research area	Selected references	Operational role in parcel sorting
Humanoid locomotion and whole-body control	Kuindersma et al. (2016); Li et al. (2026b); Tong et al. (2024); Wang et al. (2025)	Provides the basis for balance, walking, reaching, and coordinated motion in workspaces designed for people.
Manipulation, perception, and grasping	Figure AI (2025); Li et al. (2024, 2025)	Explains object detection, grasp planning, label access, parcel orientation, and stable handling under variation in shape, surface, and packaging condition.
Learning, motion retargeting, and task adaptation	Ayusawa and Yoshida (2017); Jafari et al. (2019); Radosavovic et al. (2024)	Shows how robots acquire skills, use human demonstrations, and adapt actions to changing layouts, parcel positions, and operating conditions.
Parcel sorting and robotic sorting systems	Boysen et al. (2019, 2023); Fang et al. (2025); Zhao et al. (2026)	Analyses sorter design, parcel routing, destination assignment, robot dispatching, congestion, and throughput.
Warehouse automation and RMFS optimisation	Azadeh et al. (2019); da Costa Barros and Nascimento (2021); Teck and Dewil (2022); Zhao et al. (2025)	Provides models for layout, station allocation, sequencing, robot scheduling, routing, workload balancing, and resource coordination.
Human-robot collaboration and ergonomics	Fragapane et al. (2021); Keshvarparast et al. (2025); Pasparakis et al. (2023); Winkelhaus et al. (2022)	Examines safety, fatigue, acceptance, supervision, task allocation, exception handling, and mixed human-robot work.
Queueing, simulation, and activity-based modelling	Amjath et al. (2024); Fedtke and Boysen (2017); Zhang et al. (2024)	Examines how arrivals, service times, buffers, congestion, downtime, and observed activities affect utilisation, waiting time, and effective capacity.
Cost and total cost of ownership	Ackerman (2023); Saccani et al. (2017); Zhang et al. (2023)	Examines how acquisition, integration, maintenance, software, supervision, electricity, downtime, depreciation, and residual value affect investment decisions.

manoid robots may be more suitable where existing facilities are designed for workers and full automation is uneconomic or impractical.

We consider parcel sorting as a bounded operating system. It excludes last-mile delivery, order picking, unloading, shelf handling, and general warehouse assistance. These activities may also be suitable for humanoid robots, but they involve different throughput measures, labour requirements, and operating constraints. Restricting the analysis to parcel sorting provides a baseline for comparing human and robotic resources. This is consistent with warehouse automation research, in which automated and robotised systems are evaluated through task specific throughput, utilisation, congestion, layout, and operating policies rather than specific robotic capability (see e.g. Azadeh et al., 2019; Fragapane et al., 2021; Zou et al., 2021). Table 4 defines six deployment levels, ranging from manual parcel sorting to scalable autonomous operation.

Current logistics applications remain closer to intermediate deployment levels, where robots perform selected sorting tasks while human workers continue to manage irregular parcels, operational exceptions, and recovery activities. Therefore, the economic evaluation focuses on hybrid warehouse operation rather than full warehouse automation, reflecting how work is redistributed between robots and workers in practice (Grosse et al., 2015, 2017). In this study, the numerical analysis uses operating data from two major Chinese logistics carriers, SF Express and JD Logistics, to define scenarios with different labour costs and productivity levels. These cases are used to

Table 4. Operational taxonomy of humanoid robot deployment in parcel sorting

Level	Deployment state	Description
Level 0	Manual sorting	Parcel sorting is performed by human workers. Robots are not used for physical parcel handling.
Level 1	Demonstration capability	A humanoid robot can perform isolated parcel handling actions in a controlled demonstration, without sustained warehouse throughput.
Level 2	Supervised single task handling	A humanoid robot can handle a limited range of standard parcels under close supervision, with frequent resets or human intervention.
Level 3	Shift level assisted sorting	A humanoid robot can sustain a defined sorting task over an operating window. Human workers remain responsible for exceptions, recovery and residual tasks.
Level 4	Hybrid warehouse deployment	Humanoid robots process a substantial share of standard parcels. Human workers handle irregular parcels, failed scans, damaged packaging and operational exceptions.
Level 5	Scalable autonomous labour unit	Humanoid robots operate as scalable warehouse labour units across parcel types, shifts and operating conditions with limited intervention.

examine how labour cost and worker productivity affect the robot purchase price and throughput required for economic adoption.

3.1. Data sources and model parameters

The model uses three types of input data: (*i*) human labour cost, (*ii*) humanoid robot deployment cost, and (*iii*) sorting productivity and quality. Table 5 summarises the data sources and parameter values used in the analysis. The first group measures the cost of human parcel sorting. Recruitment data from SF Express and JD Logistics are used to capture wage levels, shift patterns, employment arrangements, and employer-paid benefits. The recorded variables include monthly and hourly pay, night shift premiums, direct hire and formal employment status, social insurance, housing fund contributions, accommodation, meal allowances, piece rate pay, performance pay, and bonuses. These inputs provide a more complete estimate of annual employer labour cost than the basic wage alone. The detailed labour cost inputs are reported in Appendix A of the supplementary material. The second group describes sorting performance. It includes human throughput, shift duration, effective working time, sorting quality-related loss, seasonal capacity, the share of parcels that can be assigned to robots, and public benchmarks for automated sorting systems. These inputs determine the throughput that a robot must achieve to replace human labour economically. Worker level data on sorting errors, missed scans, parcel damage, and rework are not consistently reported in public recruitment or corporate sources. The corresponding parameters are therefore specified as reference assumptions and examined through sensitivity and uncertainty analysis. The detailed productivity and quality inputs are reported in Appendix B of the supplementary material. The third group captures the full cost of deploying a humanoid robot for parcel sorting. The Unitree G1 price is used as the base platform cost, while the analysis also includes the additional equipment and operating expenses required for warehouse use. These include end effectors, safety equipment, barcode or OCR hardware, systems integration, maintenance, software, supervision, electricity, downtime, depreciation, and residual value. This distinction is important because the purchase price alone does not represent the full cost of an operational parcel sorting system. The detailed robot cost inputs are reported in Appendix C of the supplementary material.

Table 5. Data sources and model parameters

Data group	Variables	Use in the model
Recruitment data	Monthly and hourly wages, night shift premiums, direct hire status, formal employment status, social insurance, housing fund contributions, accommodation, meal allowances, piece rate pay, performance pay, and bonuses	Estimates the annual employer labour cost of a full-time equivalent sorter
Sorting productivity and quality data	Human throughput, shift duration, effective working time, sorting quality loss, seasonal capacity, share of parcels assignable to robots, and public automated sorting benchmarks	Estimates throughput and cost per parcel for human-robot sorting
Robot price and total cost of ownership data	Base robot price, end-effectors, safety equipment, barcode or optical character recognition (OCR) hardware, systems integration, maintenance, software, supervision, electricity, downtime, depreciation, and residual value	Estimates the annual cost of deploying and operating a humanoid robot for parcel sorting

All monetary values are reported in USD. Labour costs reported in CNY are converted at an exchange rate of 6.7805 CNY per USD. Baseline labour inputs used in the numerical analysis are provided in Appendix D of the supplementary material. The second part of the cost comparison concerns the annual cost of deploying and operating a humanoid robot. Table 6 reports the cost inputs used for the Unitree G1 benchmark. Its listed price provides the base platform cost, while the remaining inputs account for the equipment, integration, operation, and support required for parcel sorting.

Table 6. Baseline cost inputs for the Unitree G1 benchmark, as of July 2026

Cost component	Baseline value
Base robot price	\$13,500
Useful life	5 years
Residual value	20% of base price
End-effector upgrade	10% of base price
Safety equipment	3% of base price
Systems integration	20% of base price
Barcode and vision system	\$737.41
Annual maintenance	8% of base price
Annual software cost	\$4,247.47
Annual supervision cost	\$3,318.34
Annual electricity cost	\$339.80
Annual downtime cost	\$5,309.34

The labour and robot cost inputs differ in how directly they can be observed or estimated. Table 7 summarises how each input is treated in the model. Inputs are classified as observed values, derived values, reference assumptions, or uncertain parameters. Publicly available values are used where possible, while derived measures and operational assumptions are stated explicitly. Parameters subject to greater uncertainty are examined through Monte Carlo simulation. This distinction clarifies the basis of each input and how uncertainty is handled in the analysis.

Table 7. Classification of model inputs

Input category	Variables	Role in the model
Observed values	Wage midpoint, recruitment benefits, shift indicators, and public robot price	Used directly where reliable public data are available
Derived values	Annual employer labour cost, quality adjusted annual human output, human sorting cost, and annual robot cost	Calculated from observed values and stated assumptions
Scenario assumptions	Human throughput, robot throughput, seasonal multipliers, share of parcels assignable to robots, and sorting quality loss	Used to define low, normal, and peak operating conditions
Uncertain parameters	Human sorting cost, effective robot output, fixed robot operating cost, and maintenance cost	Assigned probability distributions and tested through Monte Carlo simulation

3.2. *Economic model formulation*

Publicly available data on humanoid robot use in parcel sorting are limited. Wages, employment benefits, and robot purchase prices can be obtained from public sources, but operational parameters such as uptime, failed grasps, missed scans, operator requirements, and downtime costs are rarely reported. Therefore, the model considers deployment as a threshold decision under partial information. The comparison is based on the cost per successfully processed parcel rather than labour cost or robot price alone. Human labour cost per successfully processed parcel is calculated from annual employer cost, effective working time, sorting throughput, and quality-related losses. Robot cost is calculated using annual total cost of ownership, including acquisition, end-effectors, safety equipment, systems integration, maintenance, software, supervision, electricity, downtime, depreciation, and residual value (Jacob et al., 2026; Zhang et al., 2023). Low, normal, and peak operating conditions are considered separately to account for changes in parcel volume, overtime, night shift work, temporary labour use, and productivity. Economic viability is assessed using unit cost and payback period. The unit cost comparison identifies the robot throughput required to match the human cost per successfully processed parcel. The payback period shows whether the initial deployment cost can be recovered within a specified time. Together, these measures identify the combinations of robot price and quality adjusted throughput for which humanoid deployment is economically viable. Parameters that cannot be observed directly are specified as reference assumptions and examined through sensitivity and Monte Carlo analysis.

3.2.1. *Cost and capacity model*

The cost and capacity model is applied to the reference operating conditions reported for SF Express and JD Logistics. Let $j \in \{\mathrm{SF}, \mathrm{JD}\}$ denote the carrier and $s \in \{\mathrm{low}, \mathrm{normal}, \mathrm{peak}\}$ denote the demand condition. The model is evaluated separately for each carrier and demand condition. Table 8 summarises the notation.

The model first calculates the human sorting cost per successfully processed parcel. The annual employer cost of one full-time equivalent worker is defined as

$$C_{Hj} = 12W_j(1 + b_j) + O_j + T_j + RC_j, \tag{1}$$

where W_j denotes the monthly wage midpoint, b_j denotes the employer contribution rate for social insurance, housing fund contributions, and other benefits, and O_j de-

Table 8. Notation used in the cost and capacity model

Notation	Unit	Definition
Indices and operating conditions		
j	Index	Reference carrier, $j \in \{\mathrm{SF}, \mathrm{JD}\}$.
s	Index	Demand condition, $s \in \{\mathrm{low}, \mathrm{normal}, \mathrm{peak}\}$.
d	Days/year	Number of operating days per year.
Human labour cost and capacity		
W_j	USD/month	Monthly wage midpoint for carrier j.
b_j	Share	Employer contribution rate for social insurance, housing funds, and other benefits.
O_j	USD/FTE/year	Annual overtime, night shift, performance, and piece rate payments.
RC_j	USD/FTE/year	Annual recruitment and replacement cost per full-time equivalent worker.
T_j	USD/FTE/year	Annual training cost per full-time equivalent worker.
C_{Hj}	USD/FTE/year	Annual employer cost of one full-time equivalent worker for carrier j.
q_{Hj}	Parcels/hour	Hourly parcel processing rate of a worker for carrier j.
h_{Hj}	Hours/day	Daily shift length for carrier j.
u_{Hj}	Share	Effective working time share for carrier j.
n_{Hj}	Share	Productivity loss associated with night shift work.
e_{Hj}	Share	Wrong sorting rate.
m_{Hj}	Share	Missed scan rate.
a_{Hj}	Share	Parcel damage rate.
r_{Hj}	Share	Rework rate.
Q^*_{Hj}	Parcels/FTE/year	Annual number of successfully processed parcels per worker.
UC_{Hj}	USD/parcel	Human unit cost under the baseline operating condition.
Robot investment and annual cost		
P	USD	Base robot purchase price.
α_E	Share	End-effector upgrade cost as a proportion of the base robot price.
α_S	Share	Safety system cost as a proportion of the base robot price.
α_G	Share	Integration and initial setup cost as a proportion of the base robot price.
B	USD	Cost of barcode and vision equipment.
$I_R(P)$	USD	Upfront robot deployment cost at purchase price P.
ρ	Share	Residual value as a proportion of the base robot price.
L	Years	Useful economic life of the robot.
$D_R(P)$	USD/year	Annual robot depreciation.
α_M	Share/year	Annual maintenance cost as a proportion of the base robot price.
SW	USD/year	Annual software and technical support cost.
OP	USD/year	Annual cost of robot supervision and operating personnel.
EL	USD/year	Annual electricity cost.
DT	USD/year	Annual economic cost of robot downtime.
$O_R(P)$	USD/year	Annual robot operating cost.
$C_R(P)$	USD/year	Annual economic cost of robot deployment.
β_1	Year^{-1}	Coefficient of the robot purchase price in the annual robot cost function.
β_0	USD/year	Fixed component of the annual robot cost function.
Seasonal costs and robot output		
λ_s	Multiplier	Seasonal labour cost multiplier under demand condition s.
η_s	Multiplier	Seasonal human capacity multiplier under demand condition s.
UC_{Hjs}	USD/parcel	Human unit cost for carrier j under demand condition s.
q	Parcels/hour	Sustained robot throughput.
Ω_s	Hours/year equivalent	Annual robot output factor incorporating operating time, uptime, handling success, and seasonal operating conditions.
$Q_R(q, s)$	Parcels/year	Annual successfully processed robot output under demand condition s.
Economic frontiers		
T	Years	Target payback period.
$q^{UC}_{js}(P)$	Parcels/hour	Minimum robot throughput required to match the human unit cost at price P.
$P^{UC}_{js}(q)$	USD	Maximum base robot price that matches the human unit cost at throughput q.
$q^{PB}_{js}(P)$	Parcels/hour	Minimum robot throughput required to satisfy the target payback period at price P.
$P^{PB}_{js}(q)$	USD	Maximum base robot price that satisfies the target payback period at throughput q.
Hybrid workforce allocation		
D_{js}	Parcels/day	Daily parcel demand for carrier j under demand condition s.
ϕ_{js}	Share	Share of daily parcel demand for carrier j that is suitable for robotic handling under demand condition s.
$K_R(q, s)$	Parcels/robot/day	Daily processing capacity of one robot before applying the parcel-suitability constraint.
K_{Hjs}	Parcels/FTE/day	Daily processing capacity of one worker for carrier j under demand condition s.
X	Parcels/day	Number of parcels assigned to robots for the fixed carrier and demand condition.
R	Robots	Number of robots deployed.
H	FTE workers	Number of human workers deployed.
Z_{js}	USD/day	Minimum daily cost of satisfying demand for carrier j under condition s.

notes overtime, night shift, performance, and piece rate payments. The terms T_j and RC_j represent training and recruitment or replacement costs, respectively. These costs are included to account for staff turnover and repeated onboarding, particularly during peak periods when firms rely more heavily on temporary labour. Annual successfully processed parcel output per worker is defined as

$$Q^*_{Hj} = q_{Hj} h_{Hj} d u_{Hj} (1 - n_{Hj})(1 - e_{Hj})(1 - m_{Hj})(1 - a_{Hj})(1 - r_{Hj}), \quad (2)$$

where q_{Hj} denotes parcels processed per hour, h_{Hj} the shift length, d the annual number of workdays, and u_{Hj} the effective working time share. The parameter n_{Hj} captures the weighted productivity loss associated with the night-shift structure used in the reference labour profile for carrier j. Additional changes in labour cost and capacity under low, normal, and peak demand are represented separately by λ_s and η_s, respectively. The terms e_{Hj}, m_{Hj}, a_{Hj}, and r_{Hj} denote the wrong sorting, missed scan, damage, and rework rates. These losses are deducted from raw processing capacity because they reduce successfully processed output and may generate additional downstream costs. Human unit cost is then defined as

$$UC_{Hj} = \frac{C_{Hj}}{Q^*_{Hj}}. \tag{3}$$

This value provides the human labour benchmark for the cost-efficiency frontier. The robot benchmark is constructed next. Let P denote the base robot price. This price alone does not represent the full cost of deployment, which also includes equipment, integration, and initial setup costs (Jacob et al., 2026). The upfront deployment cost is defined as

$$I_R(P) = (1 + \alpha_E + \alpha_S + \alpha_G)P + B, \tag{4}$$

where α_E, α_S, and α_G denote the end-effector upgrade, safety system, and integration rates, respectively, and B denotes the cost of barcode and vision equipment. These costs are included because parcel sorting requires suitable end-effectors, safe operation around workers, integration with the sorting process, and reliable parcel identification. Annual depreciation spreads the deployable asset cost over its useful life:

$$D_R(P) = \frac{I_R(P) - \rho P}{L}, \tag{5}$$

where ρ is the residual value rate and L is the useful life. Annual robot operating cost is

$$O_R(P) = \alpha_M P + SW + OP + EL + DT, \tag{6}$$

where α_M denotes the maintenance rate, SW the software cost, OP the cost of operating personnel, EL the electricity cost, and DT the cost of downtime. These terms account for maintenance and replacement parts, software support, supervision and exception handling, electricity use, and output lost during robot downtime. The annual economic cost of robot deployment is therefore:

$$C_R(P) = D_R(P) + O_R(P) = \beta_1 P + \beta_0. \tag{7}$$

Under the baseline parameter values, this expression becomes

$$C_R(P) = 0.306P + 13{,}362.44. \tag{8}$$

At $P = \$13{,}500$, the annual economic cost is \$17,493.44 per robot. This value provides the robot cost benchmark used to construct the cost-efficiency frontier. The third part of the model accounts for seasonality. Labour cost and processing capacity vary across low, normal, and peak demand periods. Seasonal human unit cost is defined as

$$UC_{Hjs} = UC_{Hj} \frac{\lambda_s}{\eta_s}, \tag{9}$$

where λ_s denotes the seasonal labour cost multiplier and η_s the seasonal human capacity multiplier. The baseline values are $(0.95, 1.10)$ for low demand, $(1.00, 1.00)$ for normal demand, and $(1.20, 0.85)$ for peak demand. The low-demand case reflects lower labour pressure and greater available capacity, whereas the peak-demand case accounts for overtime, temporary labour, night work, congestion, and fatigue. A higher λ_s increases labour cost, while a lower η_s reduces effective capacity. Both increase the human unit cost. Annual robot output is defined as

$$Q_R(q, s) = q\Omega_s, \tag{10}$$

where q denotes sustained robot throughput and Ω_s denotes the annual successfully processed output factor. This factor accounts for operating time, uptime, handling success, and seasonal operating conditions. The share of parcels suitable for robotic handling is represented separately by ϕ_{js} in the workforce allocation model. In the baseline simulation, Ω_s is 3,677.93 under low demand, 3,432.73 under normal demand, and 2,795.23 under peak demand. Robot output is therefore based on sustained usable performance rather than maximum movement speed. Differences in Ω_s reflect changes in congestion, uptime, handling success, and other operating conditions across demand periods. The economic frontier therefore represents robot performance conditional on sufficient suitable parcel demand. The parcel-suitability limit is imposed separately in the workforce allocation model.

The fourth part of the model derives the economic frontier. Following De Kok et al. (2008); Zhang et al. (2023), the robot matches the human unit cost when

$$\frac{C_R(P)}{Q_R(q, s)} \leq UC_{Hjs}. \tag{11}$$

This condition requires the robot cost per successfully processed parcel to be no greater than the corresponding human unit cost. For a fixed robot price, the minimum throughput required to match the human unit cost is

$$q_{js}^{UC}(P) = \frac{C_R(P)}{UC_{Hjs}\Omega_s}. \tag{12}$$

For a fixed robot throughput, the maximum acceptable base robot price is

$$P_{js}^{UC}(q) = \frac{UC_{Hjs}q\Omega_s - \beta_0}{\beta_1}. \tag{13}$$

Here, $\beta_1 = 0.306$ and $\beta_0 = 13{,}362.44$. Together with the inverse expression, this equation defines the unit cost frontier. It gives the maximum acceptable robot price at a given throughput, while the inverse form gives the minimum throughput required at a given price. Matching the human unit cost represents only the minimum economic threshold. Adoption also requires the upfront deployment cost to be recovered within an acceptable payback period (Jacob et al., 2026). For a target payback period T, the payback condition is

$$\frac{I_R(P)}{UC_{Hjs}Q_R(q, s) - O_R(P)} \leq T. \tag{14}$$

The numerator represents the upfront deployment cost, while the denominator represents the annual labour cost avoided through robot deployment after deducting annual robot operating costs. A finite payback period exists only when this annual net saving is positive:

$$UC_{Hjs}Q_R(q,s) - O_R(P) > 0. \tag{15}$$

Subject to this condition, the maximum base robot price that satisfies the target payback period is

$$P_{js}^{PB}(q) = \frac{TUC_{Hjs}q\Omega_s - T(SW + OP + EL + DT) - B}{(1 + \alpha_E + \alpha_S + \alpha_G) + T\alpha_M}. \tag{16}$$

The inverse form gives the minimum robot throughput required at a fixed price:

$$q_{js}^{PB}(P) = \frac{P[(1 + \alpha_E + \alpha_S + \alpha_G) + T\alpha_M] + T(SW + OP + EL + DT) + B}{TUC_{Hjs}\Omega_s}. \tag{17}$$

The payback frontier is used as the adoption threshold because it accounts for both the upfront investment and annual operating costs. The payback frontier determines whether a robot price and throughput combination is economically acceptable. The allocation model is evaluated for selected combinations outside this frontier solely to illustrate the effects of throughput on staffing and operating cost, conditional on deployment. Such results should not be interpreted as economically justified investment decisions.

Daily robot and human capacities are obtained by converting annual output into daily values:

$$K_R(q,s) = \frac{Q_R(q,s)}{d} = \frac{q\Omega_s}{d}, \tag{18}$$

$$K_{Hjs} = \frac{\eta_s Q_{Hj}^*}{d}. \tag{19}$$

The final stage uses a mixed-integer linear programming model to determine the least-cost combination of humanoid robots and human workers required to satisfy daily parcel demand. The model is solved separately for each carrier $j \in \{\mathrm{SF}, \mathrm{JD}\}$ and demand condition $s \in \{\mathrm{low}, \mathrm{normal}, \mathrm{peak}\}$. For a fixed combination (j, s), let X denote the daily number of parcels assigned to robots. Robots process suitable parcels, while workers handle the remaining parcels and operational exceptions (Zhang et al., 2024). The problem is formulated as follows:

$$\min_{R,H,X} \quad Z_{js} = R\frac{C_R(P)}{d} + H\frac{C_{Hj}\lambda_s}{d}, \tag{20}$$

$$\text{s.t.} \quad X + HK_{Hjs} \geq D_{js}, \tag{21}$$

$$X \leq RK_R(q,s), \tag{22}$$

$$X \leq \phi_{js}D_{js}, \tag{23}$$

$$X \in \mathbb{R}_{\geq 0}, \tag{24}$$

$$R, H \in \mathbb{Z}_{\geq 0}. \tag{25}$$

The objective function in (20) minimises the total daily cost of robot deployment and human labour. Annual costs are converted to daily values using the number of operating days per year, d. Constraint (21) ensures that the combined processing capacity of robots and workers is sufficient to satisfy daily parcel demand. Constraint (22) limits the number of parcels assigned to robots by the installed robot capacity, while Constraint (23) restricts robot allocation to the share of demand suitable for robotic handling. Constraint (24) defines X as a non-negative continuous variable. Constraint (25) requires the numbers of robots and workers to be non-negative integers. Workers provide the remaining capacity for parcels beyond robot capacity, irregular parcels, and operational exceptions.

3.3. Scenario and uncertainty analysis

The seasonal scenarios capture changes in operating conditions across low, normal, and peak periods. They reflect differences in labour cost, human productivity, and the share of parcel demand available for robot processing. Peak periods include higher overtime, temporary labour, night work, congestion, and fatigue. The Monte Carlo simulation considers uncertainty within each seasonal scenario. Human unit cost, robot output, maintenance cost, and fixed operating cost may still vary across facilities and periods. For each scenario and each grid point (P, q), 2,500 parameter combinations are generated. Human unit cost is multiplied by a truncated normal variable with mean one and standard deviation 0.10. Successfully processed robot output is multiplied by a truncated normal variable with mean one and standard deviation 0.12. Fixed robot operating cost is multiplied by a truncated normal variable with mean one and standard deviation 0.10, while maintenance cost is multiplied by a truncated normal variable with mean 1 and standard deviation 0.15. For each grid point, the robustness metric is the proportion of simulated outcomes in which the three-year payback condition

$$P \leq P_{js}^{PB}(q) \tag{26}$$

is satisfied. A higher value indicates that the robot remains economically feasible across a wider range of parameter outcomes. The resulting probability is reported for each carrier, seasonal condition, robot price, and throughput level. This allows the deterministic payback frontier to be compared with its performance under uncertainty and identifies combinations that remain feasible without relying on a single set of parameter values.

4. Numerical results

This section presents the human and robot unit costs, the unit cost and payback frontiers, the maximum justifiable robot price at selected throughput levels, the optimal allocation of human and robot capacity, and the uncertainty analysis. Complete results are provided in Appendix E of the supplementary material.

4.1. Human and robot cost comparison

The analysis begins by comparing the cost per successfully processed parcel for human workers and robots. For SF Express, the annual employer cost is $14,794 per full-time equivalent worker, giving a normal season unit cost of $0.01516 per parcel after accounting for throughput, effective working time, shift patterns, and sorting losses. For JD Logistics, the annual employer cost is $15,628.79, while the lower assumed worker throughput results in a higher unit cost of $0.02102 per parcel. At the Unitree G1 price of $13,500, the annual robot cost is $17,493.44, including depreciation, maintenance, software, supervision, electricity, and downtime. This is higher than the annual cost of one worker in both cases. However, annual cost alone is not sufficient for comparison because robots and workers may process different parcel volumes. Economic viability therefore depends on the cost per successfully processed parcel and, in particular, on whether the robot can sustain enough throughput to recover its purchase and operating costs. Table 9 reports the throughput required at the Unitree G1 price. The unit cost threshold is the throughput at which robot and human unit costs are equal. The three-year payback threshold is higher because the initial deployment cost must also be recovered within three years.

Table 9. Required robot throughput at the Unitree G1 benchmark price of $13,500

Carrier	Season	Human unit cost (USD/parcel)	Unit cost threshold (pph)	Three-year payback threshold (pph)
SF Express	Low	0.01309	363.3	426.2
SF Express	Normal	0.01516	336.1	394.4
SF Express	Peak	0.02140	292.4	343.1
JD Logistics	Low	0.01815	262.0	307.4
JD Logistics	Normal	0.02102	242.4	284.5
JD Logistics	Peak	0.02968	210.9	247.4

Under normal season conditions, the three-year payback threshold is 394.4 parcels per hour for SF Express and 284.5 parcels per hour for JD Logistics. The lower threshold for JD Logistics follows from its higher human unit cost. As human sorting becomes more expensive, less robot throughput is required to recover the deployment cost within three years.

Figure 1 presents the cost-efficiency results. Figure 1(a) shows the three-year payback frontiers under normal season conditions, with the horizontal dotted line indicating the Unitree G1 benchmark price. Figure 1(b) reports the throughput required at this price across seasonal conditions. Figures 1(c) and (d) show the feasible price and throughput combinations for SF Express and JD Logistics under normal season conditions. The grey areas indicate combinations that satisfy the three-year payback condition. The required throughput is highest in the low season and lowest in the peak season. For SF Express, the threshold decreases from 426.2 parcels per hour in the low season to 343.1 parcels per hour in the peak season. For JD Logistics, it decreases from 307.4 to 247.4 parcels per hour. Peak periods raise the effective cost of human sorting through overtime, temporary labour, night work, congestion, and lower effective capacity. Therefore, the robot requires less throughput to satisfy the payback condition. Table 10 reports the maximum acceptable base robot price under a three-year payback criterion for representative robot throughput levels of 300, 400, 500, and 600 parcels per hour (pph) during the normal demand condition.

The maximum acceptable robot price increases with sustained throughput because the annual robot cost is distributed across more successfully processed parcels. For

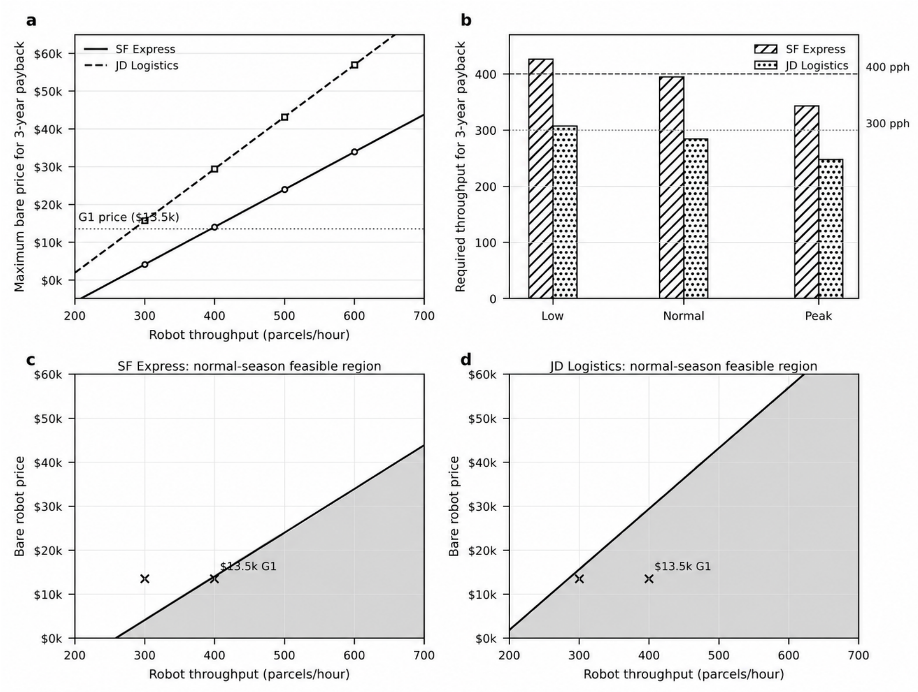

Figure 1. Cost-efficiency frontier

Table 10. Maximum acceptable robot price under a three-year payback criterion in the normal season

Carrier	300 pph	400 pph	500 pph	600 pph
SF Express	\$4,112	\$14,056	\$24,000	\$33,944
JD Logistics	\$15,644	\$29,432	\$43,220	\$57,009

SF Express, the Unitree G1 price is not justified at 300 parcels per hour but becomes feasible at approximately 400 parcels per hour. JD Logistics supports a higher price at each throughput level because its human unit cost is higher. The next analysis considers the effects of robot deployment on staffing and daily operating cost. Figure 2 reports the hybrid deployment results at the Unitree G1 benchmark price and a sustained throughput of 400 parcels per hour.

Figure 2(a) shows the number of robots required as daily demand increases, while Figure 2(b) shows the corresponding number of human workers. Figure 2(c) compares the daily costs of all human and hybrid operations under normal season demand growth. Figure 2(d) reports the cost savings across the two cases and seasonal conditions.

4.2. Hybrid staffing

The hybrid allocation model determines the numbers of robots and human workers required at the Unitree G1 benchmark price. Daily parcel demand is set at 50,000 in the low season, 100,000 in the normal season, and 200,000 in the peak season. The share of demand assignable to robots is limited to 75%, 70%, and 60%, respectively, leaving irregular parcels and exceptions to human workers. Table 11 reports the optimal numbers of robots and human workers for each season and robot throughput

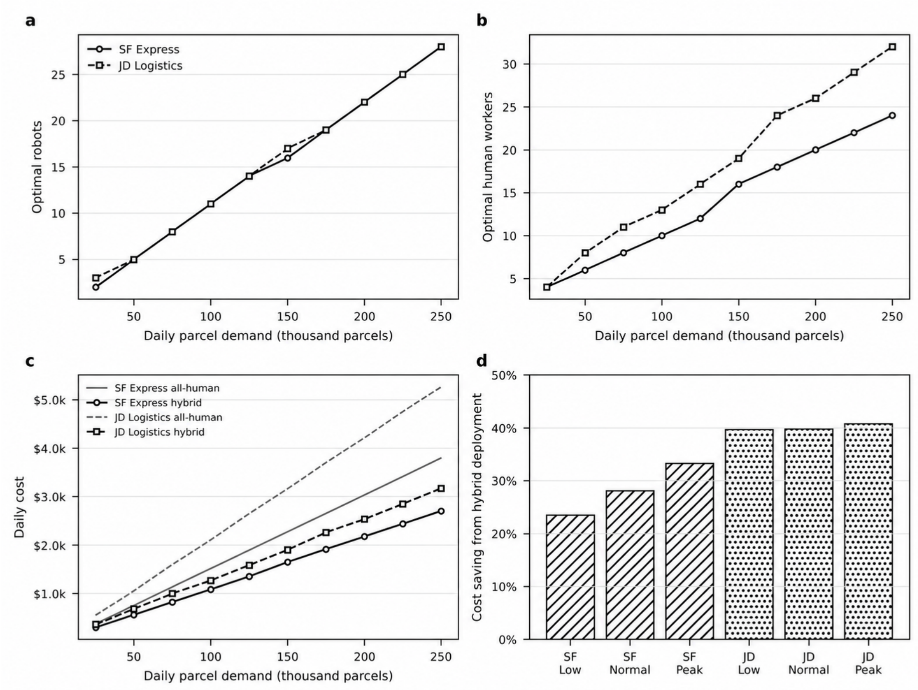

Figure 2. Hybrid deployment outcomes

level.

Table 11. Hybrid staffing outcomes at the Unitree G1 benchmark price, with three-year payback feasibility

Carrier	Season	Throughput (pph)	Robots	Humans	Robot share (%)	Hybrid cost/day ($)	Savings (%)	Three-year payback feasible
SF Express	Low	300	7	5	66.0	618	8.6	No
SF Express	Low	400	6	4	75.0	517	23.5	No
SF Express	Low	500	5	4	75.0	461	31.8	Yes
SF Express	Normal	300	14	11	66.0	1,307	13.9	No
SF Express	Normal	400	11	10	69.2	1,091	28.1	Yes
SF Express	Normal	500	9	10	70.0	979	35.5	Yes
SF Express	Peak	300	27	31	60.0	3,278	24.2	No
SF Express	Peak	400	20	31	59.7	2,885	33.3	Yes
SF Express	Peak	500	16	31	59.7	2,661	38.5	Yes
JD Logistics	Low	300	8	5	75.0	686	27.9	No
JD Logistics	Low	400	6	5	75.0	574	39.7	Yes
JD Logistics	Low	500	5	5	75.0	518	45.5	Yes
JD Logistics	Normal	300	15	13	70.0	1,492	29.1	Yes
JD Logistics	Normal	400	11	13	69.2	1,268	39.7	Yes
JD Logistics	Normal	500	9	13	70.0	1,156	45.1	Yes
JD Logistics	Peak	300	27	40	60.0	3,918	34.2	Yes
JD Logistics	Peak	400	20	40	59.7	3,526	40.8	Yes
JD Logistics	Peak	500	16	40	59.7	3,302	44.5	Yes

At a sustained throughput of 400 parcels per hour in the normal season, the optimal allocation is 11 robots and 10 human workers for SF Express, reducing daily cost by 28.1%. For JD Logistics, the corresponding allocation is 11 robots and 13 human workers, reducing daily cost by 39.7%. Under peak-season conditions, savings increase to 33.3% for SF Express and 40.8% for JD Logistics. The analysis next examines whether these results remain valid under parameter uncertainty. Figure 3 reports the probability of satisfying the three-year payback condition across the simulated parameter values.

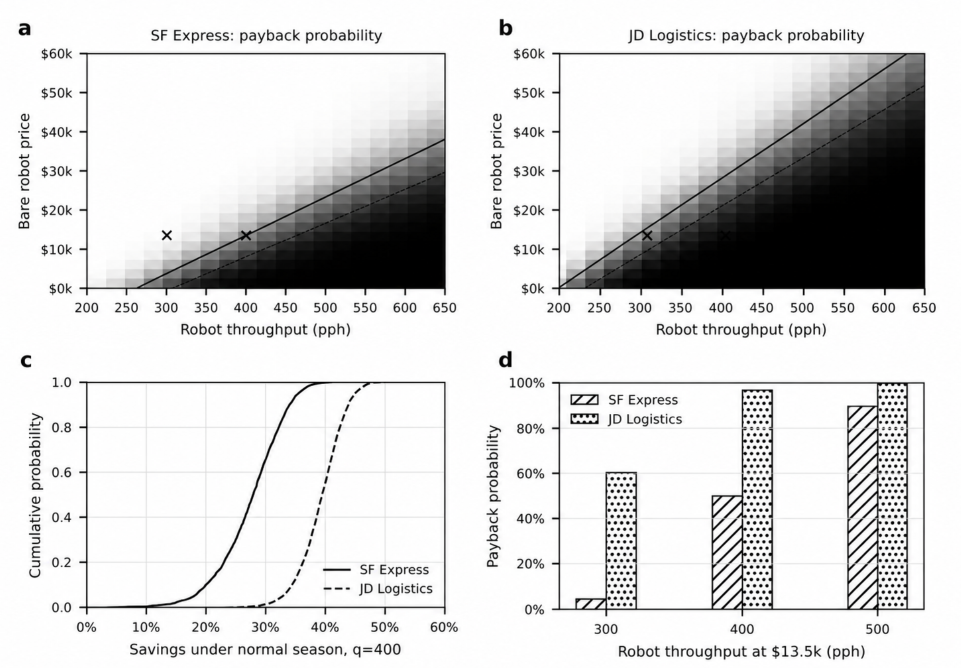

Figure 3. Payback results under parameter uncertainty

Figures 3(a) and (b) show the probability of satisfying the three-year payback condition across robot price and throughput combinations for SF Express and JD Logistics under normal season conditions. Figure 3(c) shows the cumulative distribution of cost savings at the Unitree G1 benchmark price and a throughput of 400 parcels per hour. Figure 3(d) reports the payback probability at the $13,500 benchmark price for selected throughput levels. Darker areas indicate a higher probability of satisfying the payback condition.

The allocation results support partial rather than full labour replacement. Robots are used only for the share of parcels assumed to be suitable for automated handling, while human workers process the remaining parcels and manage exceptions. This division is maintained across both cases and all seasonal conditions. Standard, scannable, and physically manageable parcels can be assigned to robots, whereas irregular parcels, damaged packaging, unreadable labels, and unstable items continue to require human judgement and recovery. Therefore, the economic role of the robot is to reduce the cost of repetitive handling rather than remove the need for human labour. Higher robot throughput improves the hybrid solution mainly by reducing the number of robots required. Under normal season conditions, increasing throughput from 300 to 500 parcels per hour reduces the robot count from 14 to nine for SF Express and from 15 to nine for JD Logistics. Human staffing changes little because the proportion of demand assignable to robots remains fixed. Throughput affects the capital intensity of deployment, while the assignable workload share determines how far human staffing can be reduced. A faster robot does not remove the residual workload or the need for exception handling.

The uncertainty analysis shows that economic viability depends on the margin between the robot and human unit costs. At the Unitree G1 price and 300 parcels per hour, the SF Express case has a low probability of satisfying the three-year payback condition, while JD Logistics is closer to the feasible region because its human unit cost is higher. At 400 parcels per hour, the probability increases in both cases, although JD Logistics has a larger margin. The distribution of cost savings shows the same result. Savings are higher and less sensitive to parameter variation when human unit cost is higher and robot output is sufficient to spread fixed costs across a larger parcel volume. These results should not be interpreted as a general comparison between SF Express and JD Logistics. They follow from the labour costs, worker throughput, and other assumptions used for the two reference cases. More generally, a credible deployment case requires both sufficient robot throughput and a sufficiently large share of suitable parcels. Throughput determines how many robots are needed, while task coverage determines how much human work can be reassigned.

5. Discussion and implications

The numerical results show that the economic viability of humanoid robots in parcel sorting depends on sustained throughput, deployment cost, human unit cost, and the share of parcels that can be processed reliably. Demonstrated task capability alone is insufficient. A robot must sustain parcel handling performance under congestion, downtime, parcel variation, and intervention requirements while processing enough parcels to recover its acquisition and operating costs (Shi et al., 2025; Zhang et al., 2023). Parcel sorting provides a suitable application because it combines repetitive handling with frequent operational variability (Jaghbeer et al., 2020). Standard parcels can be processed efficiently, whereas damaged labels, irregular packaging, and other exceptions continue to require human judgement (Shi et al., 2025). The results therefore support hybrid deployment, where robots undertake repetitive handling and human workers manage the remaining workload and exceptions.

5.1. Economic interpretation of the cost-efficiency frontier

The cost-efficiency frontier is shaped by robot price, sustained throughput, and human unit cost. Robot price determines the investment to be recovered, while throughput determines the number of successfully processed parcels over which this cost is distributed. Human unit cost provides the benchmark against which robot deployment is assessed. A low purchase price is therefore insufficient if throughput is low, downtime is frequent, or substantial human support is required. Conversely, a more expensive robot may still be viable if it sustains higher output and requires less intervention. This interpretation is consistent with total cost of ownership and time-driven costing approaches (Klar et al., 2025; Zhang et al., 2023).

Throughput is especially important because depreciation, integration, software, maintenance, supervision, and downtime costs are incurred even when output is low. As throughput increases, these costs are spread across more parcels, reducing robot unit cost and increasing the maximum price that can be justified. The results show that increasing sustained throughput from approximately 300 to 400–500 parcels per hour can move deployment from an uneconomic position into a viable hybrid range. This is consistent with warehouse operations research, where realised capacity and utilisation are central determinants of system performance (Boysen and De Koster, 2025; Zou et al., 2021).

Under normal season conditions, increasing throughput from 300 to 400 parcels per hour raises the maximum justifiable robot price from \$4,112 to \$14,056 for SF Express and from \$15,644 to \$29,432 for JD Logistics. This difference reflects the modelled human unit costs rather than a general difference between the two carriers. A higher human unit cost increases the economic value of the same robot price and throughput. Seasonality changes the frontier by altering the effective cost of human sorting. Overtime, temporary labour, night work, congestion, and fatigue raise human unit cost during peak periods, reducing the throughput required for a three-year payback. Early deployment may therefore be more attractive in labour-constrained stations, night shifts, recurrent peak periods, and operations with a high proportion of standard parcels. These findings also indicate that annual averages can conceal favourable deployment conditions. Economic evaluation should be conducted at the level of the sorting station and operating period, where labour cost, parcel mix, task coverage, demand pressure, and intervention requirements can be observed directly. Humanoid robot adoption should therefore be assessed as a local process decision rather than a uniform facility wide replacement strategy.

5.2. Hybrid deployment and work design

The hybrid allocation results show that humanoid robots are more likely to complement human workers than replace them entirely. Robots are assigned to the share of the parcel workload that can be processed reliably, while human workers keep responsibility for residual parcels, supervision, and exception handling. Standard cartons with readable labels and stable handling requirements are suitable for repeated robotic processing. Irregular parcels, damaged packaging, unreadable labels, unstable items, and recovery tasks continue to require human judgement. This division of work is consistent with research that considers human and robot operations as a joint allocation and coordination problem (Gombolay et al., 2018; Lasota et al., 2017).

Economic performance depends not only on robot throughput but also on the amount of suitable work available. Assigning too little work to robots limits utilisation and spreads their fixed costs across too few parcels. Assigning tasks beyond their reliable capability can increase failed scans, rework, resets, and delays. The cost minimising allocation assigns robots to stable and repetitive tasks and directs human labour towards variability, recovery, supervision, and judgement. At 400 parcels per hour in the normal season, the model selects 11 robots and 10 human workers for SF Express, reducing daily cost by 28.1%. For JD Logistics, it selects 11 robots and 13 human workers, reducing daily cost by 39.7%. Under peak-season conditions, the corresponding savings increase to 33.3% and 40.8%. These results show that robots reduce the amount of repetitive human work without removing the need for human labour. Higher robot throughput mainly reduces the number of robots required. Under normal season conditions, increasing throughput from 300 to 500 parcels per hour reduces the robot count from 14 to 9 for SF Express and from 15 to 9 for JD Logistics. Human staffing changes little because the share of demand assignable to robots remains constrained. Therefore, throughput determines the capital intensity of deployment, while task coverage determines the extent to which human labour can be reduced.

The assignable workload share is an important practical limit. Restricting this share prevents the model from assigning the entire parcel flow to robots and reflects the continuing need for human handling of unsuitable parcels and exceptions. Partial task coverage can still be economically viable when robots process a sufficiently large standard workload at a unit cost below the human benchmark. Broader deployment would require stronger evidence on parcel variety, uptime, intervention frequency, and exception recovery. Table 12 adds an economic interpretation to the operational levels introduced in Table 4. The levels are not determined by payback alone. They also require the corresponding task coverage, operating duration, and intervention performance.

Table 12. Economic and operational conditions for humanoid robot deployment

Level	Economic condition	Work allocation	Model evidence	Deployment stage
Level 3	The robot can perform selected tasks, but the price and throughput do not yet support a reliable three-year payback.	Robots are used for limited supervised tasks, while workers handle most parcels, resets, and exceptions.	At \$13,500 and 300 parcels per hour, SF Express remains outside the normal season payback frontier. JD Logistics satisfies the deterministic threshold but remains sensitive to parameter uncertainty.	Pilot use
Level 4	The robot satisfies the three-year payback condition for a defined parcel flow and operating period.	Robots process standard parcels, while workers handle irregular parcels, damaged labels, unstable items, supervision, and recovery.	At 400 parcels per hour in the normal season, the model selects 11 robots and 10 workers for SF Express and 11 robots and 13 workers for JD Logistics. Daily savings are 28.1% and 39.7%, respectively.	Selected station or shift deployment
Level 5	The robot remains economically viable across a wider range of parcel flows, seasons, shifts, and labour cost conditions.	Robots process a broader parcel mix with less frequent human intervention.	The model limits the robot share to 60%–75% and retains human exception handling. The present results therefore do not establish this level.	Wider deployment across facilities

As stated in Table 12, Level 3 refers to supervised use for selected tasks without a reliable economic case. Level 4 refers to hybrid deployment at a specific station or shift, where the robot satisfies the payback condition but workers continue to handle residual parcels and exceptions. Level 5 refers to wider deployment across facilities and operating conditions, supported by evidence on sustained throughput, uptime, parcel coverage, intervention requirements, and payback under uncertainty.

5.3. Economic readiness under uncertainty

The uncertainty analysis shows that payback is mainly determined by human unit cost and sustained robot throughput. Higher human unit cost strengthens the case for deployment, while higher throughput reduces robot cost per successfully processed parcel. At lower throughput levels, payback becomes more sensitive to changes in robot output, maintenance, and other fixed operating costs, especially where human sorting is already efficient. Therefore, a robust deployment decision should remain economically viable under plausible variation in these parameters.

The results show that technical capability alone is not sufficient for deployment. Humanoid robots become economically attractive only when enough suitable parcels are available, the required throughput can be maintained with limited intervention, and the robot price remains within the viable range. Under the assumptions considered, the Unitree G1 benchmark is close to the transition from supervised task use to controlled hybrid deployment. Wider deployment would require evidence of stable performance across facilities, seasons, shifts, and parcel mixes. At around 300 pph, deployment is generally limited to supervised or case specific applications, particularly when human unit cost is relatively low or task coverage is narrow. At approximately 400–500 parcels per hour, the case for controlled hybrid deployment becomes stronger, provided that sufficient workload can be assigned to the robot and intervention remains limited. Reaching Level 5 would require stable performance across different parcel types, seasons, shifts, and facilities.

6. Conclusions

This study has examined the conditions under which humanoid robots can become economically viable in parcel sorting. The results show that economic readiness depends not only on technical capability, but also on whether the robot can sustain sufficient throughput, handle a sufficient volume of suitable parcels, and remain within the payback frontier under uncertainty. The cost-efficiency frontier shows that robot price, sustained throughput, human unit cost, assignable workload share, and support requirements must be considered together. Purchase price alone is insufficient. Integration, maintenance, software, supervision, electricity, downtime, and recovery determine the total cost of robotic capacity, while throughput determines how widely these costs are spread across successfully processed parcels.

The numerical results indicate that early deployment should be selective. At approximately 300 parcels per hour, the economic case remains uncertain when human unit cost is relatively low. At 400–500 parcels per hour, hybrid deployment becomes more practical, especially when robots can handle a high volume of standard parcels with limited human support. These values are carrier specific and depend on labour productivity, seasonal demand, parcel mix, and operating requirements. The allocation results also show that humanoid robots are unlikely to replace human workers completely in the near term. Robots are better suited to standard and repetitive handling tasks, while workers remain responsible for irregular parcels, failed scans, damaged packaging, supervision, and recovery. The main benefit is therefore a different division of work between robots and human workers.

The computational results reveal that deployment should be assessed for a specific station, shift, and parcel workload rather than through isolated demonstrations. The strongest pilot cases are operations with high labour pressure, recurrent peaks, night work, high staff turnover, and a large share of standard parcels. Evaluation should focus on sustained throughput, uptime, failed handling attempts, reset frequency, scan success, intervention time, downtime, rework, and cost per successfully processed parcel. These measures provide a more reliable basis for procurement than peak speed alone. The deployment levels also separate technical capability from operational readiness. Level 3 represents supervised task performance without a reliable payback case, while Level 4 represents hybrid deployment in a defined operating setting. Level 5 would require consistent performance across facilities, seasons, parcel mixes, and operating conditions. Under the assumptions considered, the Unitree G1 benchmark lies close to the transition between Levels 3 and 4.

Future research should test the model using data from real pilot deployments. The most important next step is to measure sustained throughput, uptime, intervention frequency, recovery time, parcel suitability, and human support requirements over complete shifts rather than controlled demonstrations. The model should also account for congestion, workstation layout, parcel mix, learning effects, worker acceptance, and competition from fixed or mobile automation. These extensions would allow the cost-efficiency frontier to be estimated for individual stations and operating periods, providing a practical basis for deployment, procurement, and capacity decisions.

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The authors report that there are no competing interests to declare.

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Data availability statement

The data used in this study are derived from publicly available sources and the assumptions reported in the manuscript and supplementary materials. Data are available in the GitHub repository at https://github.com/Bao-tong/Data.git.

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